

ST5209/X Assignment 5

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Set up

1. Make sure you have the following installed on your system: $\text{\LaTeX}$, R4.2.2+, RStudio 2023.12+, and Quarto 1.3.450+.
2. Pull changes from the course [repo](#).

Submission

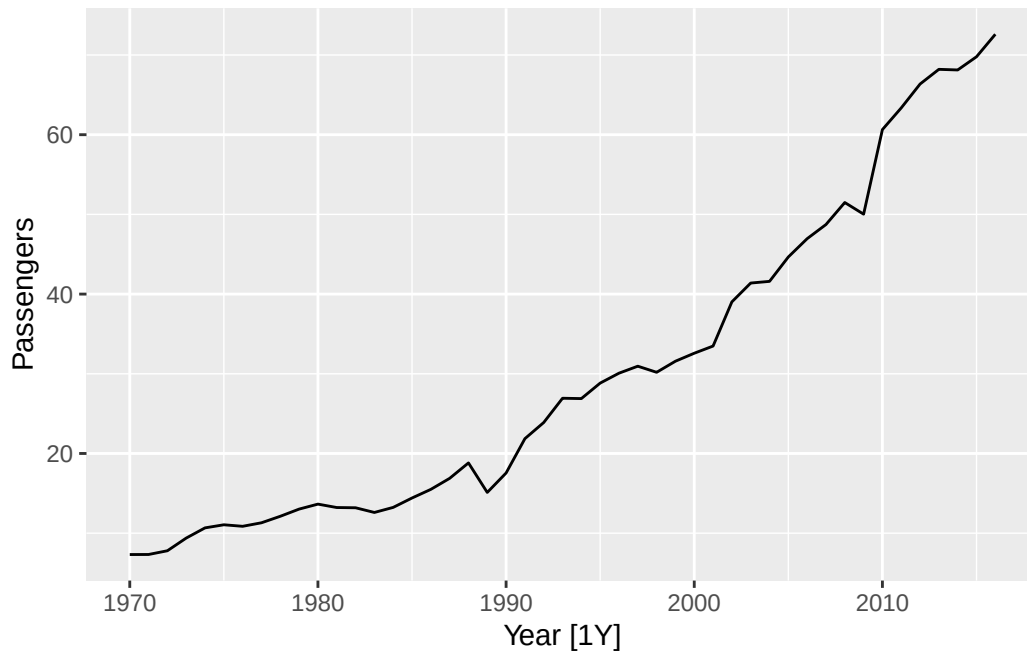
1. Render the document to get a .pdf printout.
2. Submit both the .qmd and .pdf files to Canvas.

1. Modeling with ARIMA (Q9.7 in Hyndman & Athanasopoulos)

Consider `aus_airpassengers`, the total number of passengers (in millions) from Australian air carriers for the period 1970-2016.

```
aus_airpassengers |>  
  autoplot()
```

Plot variable not specified, automatically selected ``vars = Passengers``



- a. Use `ARIMA()` to find an appropriate ARIMA model. What model was selected. Check that the residuals look like white noise. Plot forecasts for the next 10 periods.

```
arima_fit <- aus_airpassengers |>
  model(ARIMA(Passengers))

report(arima_fit)
```

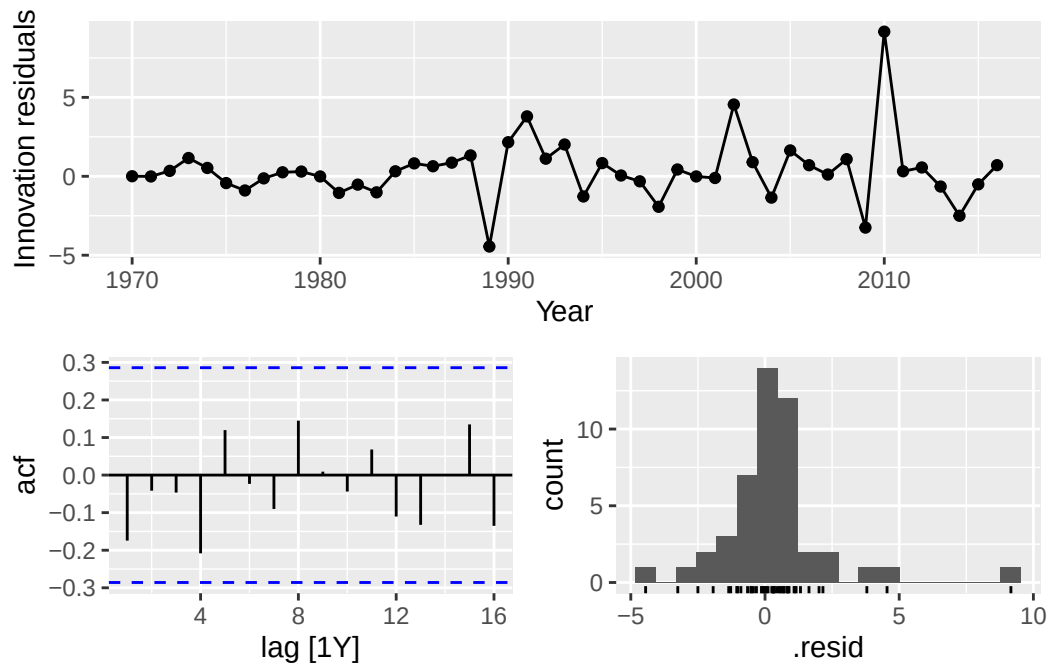
```
Series: Passengers
Model: ARIMA(0,2,1)
```

```
Coefficients:
      ma1
    -0.8963
s.e.    0.0594
```

```
sigma^2 estimated as 4.308:  log likelihood=-97.02
AIC=198.04  AICc=198.32  BIC=201.65
```

```
arima_fit |> gg_tsresiduals()
```

```
Warning: `gg_tsresiduals()` was deprecated in feasts 0.4.2.
i Please use `ggtime::gg_tsresiduals()` instead.
```

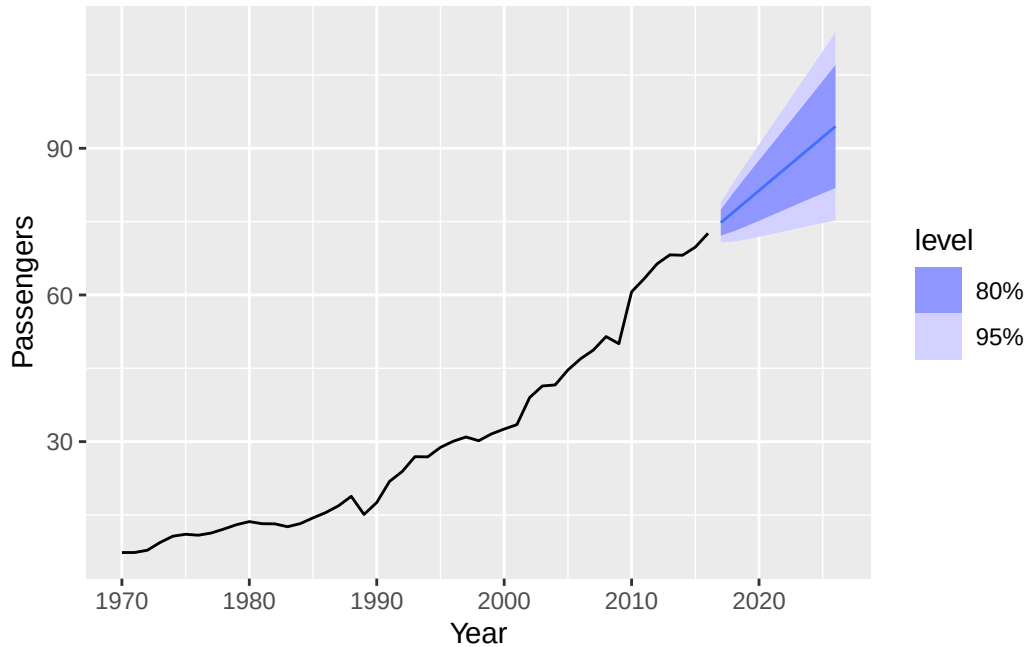


```
augment(arima_fit) |>
  features(.innov, lbjung_box, lag = 10)
```

```
# A tibble: 1 x 3
  .model      lb_stat lb_pvalue
  <chr>      <dbl>   <dbl>
1 ARIMA(Passengers) 6.70    0.754
```

```
fc <- arima_fit |> forecast(h = 10)
```

```
fc |>
  autoplot(aus_airpassengers)
```



The automatic `ARIMA()` function selected an $ARIMA(0,2,1)$ model for the series. Residual diagnostics, including the ACF plot and Ljung-Box test, confirm the residuals resemble white noise, indicating a good fit, and the 10-period forecast shows a continuation of the upward trend with widening prediction intervals.

- b. Write the model in terms of the backshift operator.

The $ARIMA(0,2,1)$ model is written as: $[(1 - B)^2 y_t = (1 - \theta_1 B) \varepsilon_t]$

Using $\theta_1 = -0.8963$,

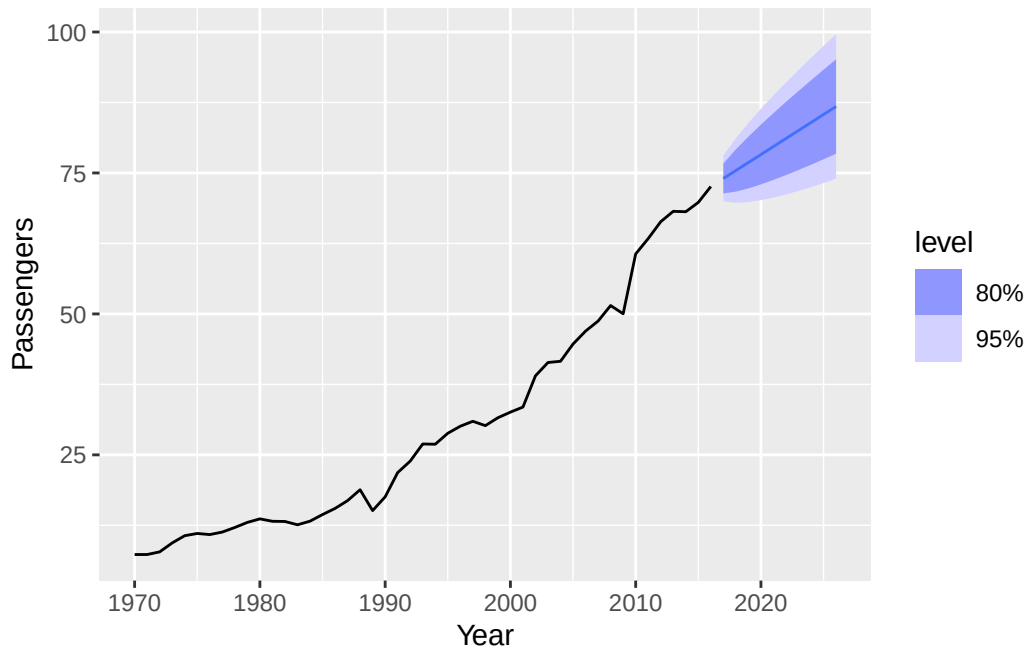
$$(1 - B)^2 y_t = (1 - 0.8963B) \varepsilon_t.$$

Where $(1 - B)^2$ applies second-order differencing and $(1 - \theta_1 B)$ is the $MA(1)$ term.

- c. Plot forecasts from an $ARIMA(0,1,0)$ model with drift and compare these to part a.

```
fit_c <- aus_airpassengers |>
  model(ARIMA(Passengers ~ 1 + pdq(0,1,0)))

fit_c |> forecast(h=10) |>
  autoplot(aus_airpassengers)
```



ARIMA(0,1,0) with drift is a random walk with a constant upward drift; forecasts are linear but less flexible than part (a), producing a straight-line projection.

- d. Plot forecasts from an ARIMA(2,1,2) model with drift and compare these to part a. Remove the constant and see what happens.

```
fit_d <- aus_airpassengers |>
  model(
    with_const = ARIMA(Passengers ~ 1 + pdq(2,1,2)),
    without_const = ARIMA(Passengers ~ 0 + pdq(2,1,2))
  )
```

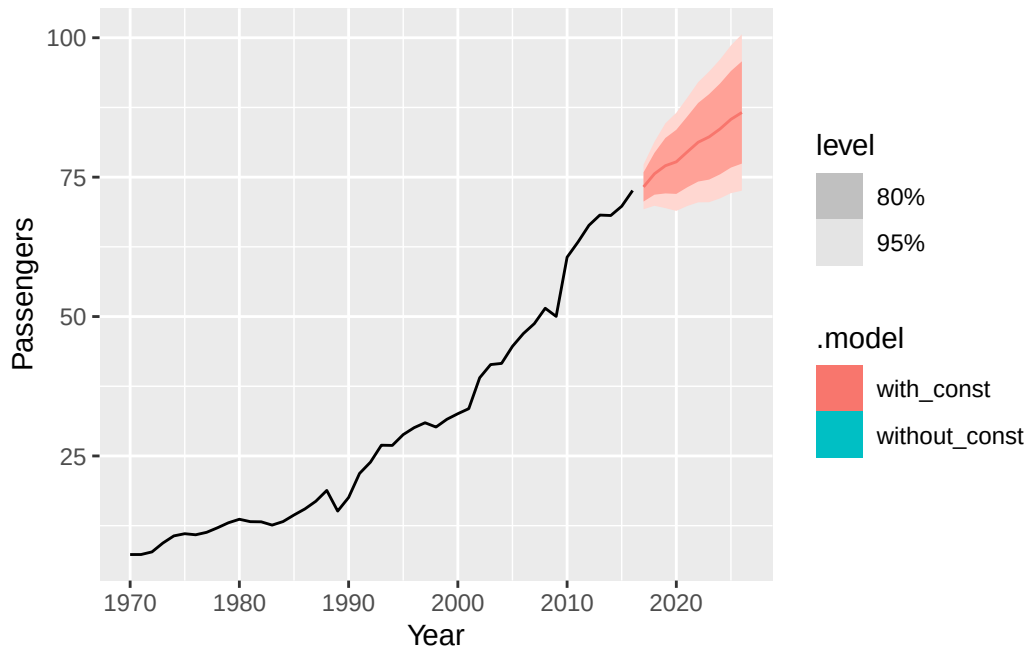
Warning: 1 error encountered for without_const
[1] non-stationary AR part from CSS

```
fit_d |> forecast(h=10) |>
  autoplot(aus_airpassengers)
```

Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning -Inf

Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning -Inf

Warning: Removed 10 rows containing missing values or values outside the scale range (`geom\_line()`).



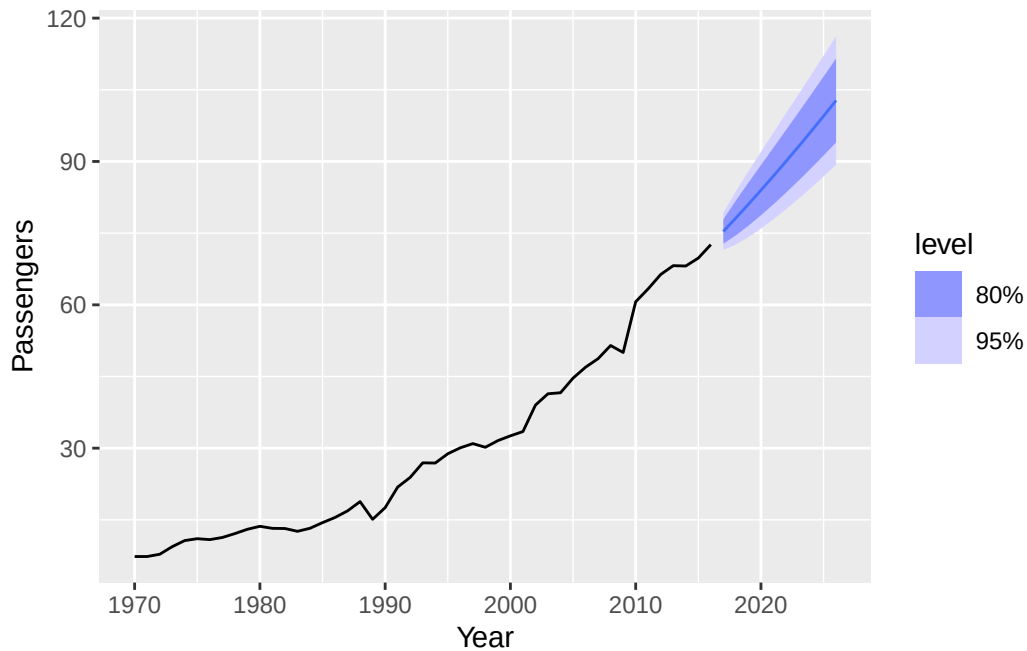
ARIMA(2,1,2) with drift produces forecasts similar to the auto-selected ARIMA(0,2,1) from part (a). Removing the constant causes the model to fail with a non-stationary AR error, highlighting that the constant is necessary to stabilise the ARIMA(2,1,2) fit.

- e. Plot forecasts from an ARIMA(0,2,1) model with a constant. What happens?

```
fit_e <- aus_airpassengers |>
  model(ARIMA(Passengers ~ 1 + pdq(0,2,1)))
```

Warning: Model specification induces a quadratic or higher order polynomial trend. This is generally discouraged, consider removing the constant or reducing the number of

```
fit_e |> forecast(h=10) |>
  autoplot(aus_airpassengers)
```



Adding a constant to an $ARIMA(0, 2, 1)$ (i.e., $d = 2$) introduces a quadratic trend into the forecasts — the predictions curve upward increasingly steeply, which is usually unrealistic for long horizons.

2. Seasonal ARIMA

- a. Load `diabetes.rds` from the directory `_data/cleaned`.

```
diabetes <- readRDS("../_data/cleaned/diabetes.rds")
diabetes
```

```
# A tsibble: 204 x 3 [1M]
  Month TotalC Cost
  <mt>    <dbl> <dbl>
1 1991 Jul  3526591 3.53
2 1991 Aug  3180891 3.18
3 1991 Sep  3252221 3.25
4 1991 Oct  3611003 3.61
5 1991 Nov  3565869 3.57
6 1991 Dec  4306371 4.31
7 1992 Jan  5088335 5.09
8 1992 Feb  2814520 2.81
9 1992 Mar  2985811 2.99
10 1992 Apr  3204780 3.20
```

```
# i 194 more rows
```

- b. Perform the following transformation of the column **Cost**: Apply a log transform followed by a seasonal difference. Label the resulting time series **Y**.

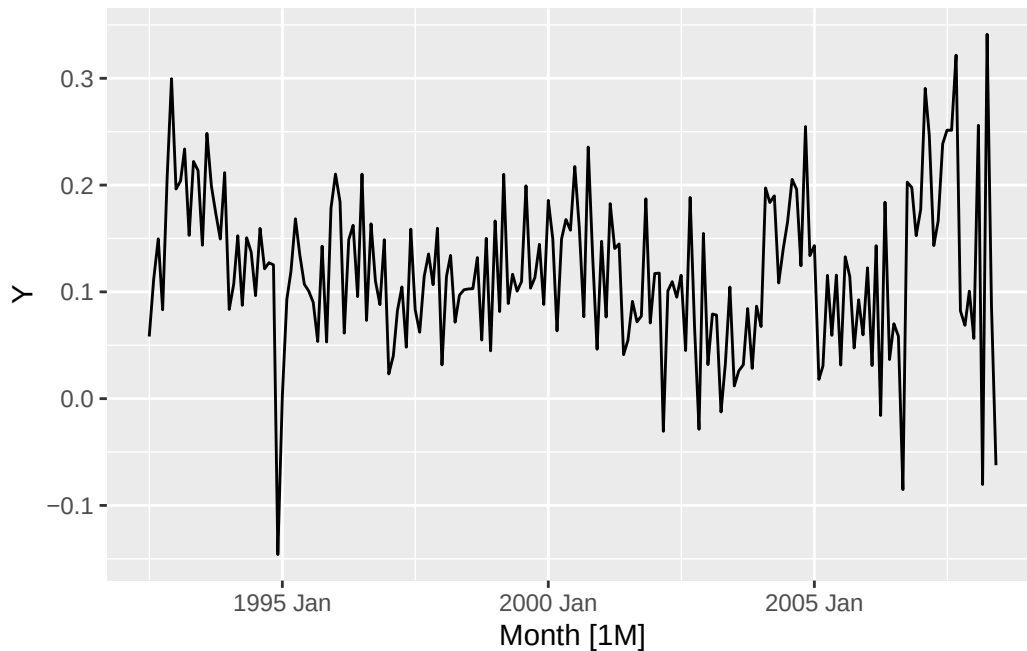
```
diabetes <- diabetes |>
  mutate(log_cost = log(Cost))

Y <- diabetes |>
  mutate(Y = difference(log(Cost), lag = 12)) |>
  filter(!is.na(Y))

Y
```

```
# A tsibble: 192 x 5 [1M]
  Month   TotalC Cost log_cost     Y
  <mth>   <dbl> <dbl>   <dbl> <dbl>
1 1992 Jul 3737851.  3.74    1.32 0.0582
2 1992 Aug 3558776.  3.56    1.27 0.112
3 1992 Sep 3777202.  3.78    1.33 0.150
4 1992 Oct 3924490.  3.92    1.37 0.0833
5 1992 Nov 4386531.  4.39    1.48 0.207
6 1992 Dec 5810549.  5.81    1.76 0.300
7 1993 Jan 6192068.  6.19    1.82 0.196
8 1993 Feb 3450857.  3.45    1.24 0.204
9 1993 Mar 3772307.  3.77    1.33 0.234
10 1993 Apr 3734303.  3.73    1.32 0.153
# i 182 more rows
```

```
Y |> autoplot(Y)
```

- c. Apply the KPSS test to Y . What is its p-value? What can you conclude about Y ?

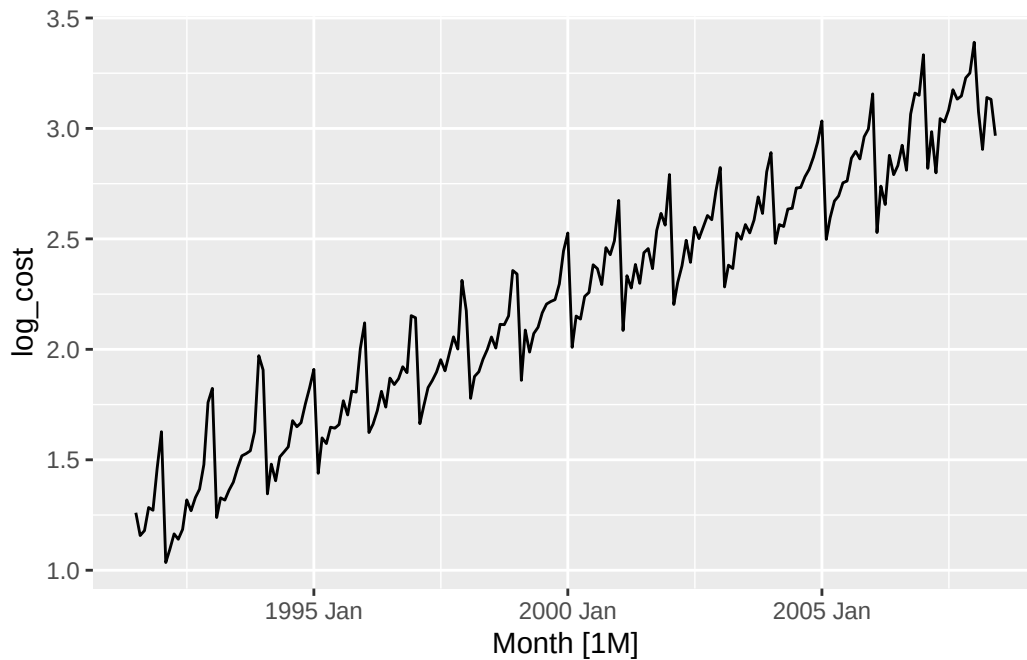
```
Y |> features(Y, unitroot_kpss)
```

```
# A tibble: 1 x 2
  kpss_stat kpss_pvalue
    <dbl>      <dbl>
1    0.209        0.1
```

The KPSS test has H_0 of stationarity, so a p-value of 0.1 means we fail to reject H_0 — Y is stationary. After applying the log and seasonal difference, the series has no remaining trend or seasonality.

- d. Make a time plot of $\log(\text{Cost})$. Why does the trend disappear when we consider Y ?

```
diabetes |>
  autoplot(log_cost)
```



log(Cost) still exhibits an upward trend and seasonal pattern. Taking the seasonal difference removes both by subtracting each value from the same month one year prior, leaving Y stationary.

- e. Fit an ARIMA model to $\log(\text{Cost})$ and report the order of the fitted model.

```
fit <- diabetes |>
  model(ARIMA(log(Cost)))

report(fit)
```

```
Series: Cost
Model: ARIMA(2,0,2)(1,1,1)[12] w/ drift
Transformation: log(Cost)
```

Coefficients:

	ar1	ar2	ma1	ma2	sar1	sma1	constant
	0.7211	0.1081	-0.6809	0.1198	0.2236	-0.8464	0.0151
s.e.	0.2364	0.2064	0.2316	0.1752	0.1112	0.0835	0.0004

```
sigma^2 estimated as 0.003595: log likelihood=266.01
AIC=-516.02 AICc=-515.24 BIC=-489.96
```

- f. How many fitted parameters are there in the model?

The fitted model is $ARIMA(2,0,2)(1,1,1)[12]$ with drift, which has 2 AR coefficients ($ar1$, $ar2$), 2 MA coefficients ($ma1$, $ma2$), 1 seasonal MA coefficient ($sma1$), 1 seasonal AR coefficient ($sar1$) and 1 drift constant. This gives a total of 7 fitted parameters.

3. Model selection

- a. What are the null hypothesis and assumptions of the ADF test?

The ADF test has null hypothesis that the series contains a unit root, so it is non-stationary. The alternative is stationarity.

It assumes the series can be represented by an autoregressive model with enough lagged differences included to remove serial correlation in the errors, and that the deterministic part of the model has been specified correctly.

- b. Is it possible for both the ADF and KPSS test applied to a dataset to have large p-values? Explain why or why not.

Yes. This is possible because the ADF and KPSS tests have different null hypotheses. ADF tests non-stationarity as the null, while KPSS tests stationarity as the null. If both p-values are large, the evidence is inconclusive: neither non-stationarity nor stationarity is strongly rejected - the data is simply not informative enough to distinguish between stationary and non-stationary behaviour, i.e., the series lies in an ambiguous borderline region.

- c. What are the AIC and AICc penalties for the model fitted in Q2?

For the fitted model ($ARIMA(2,0,2)(1,1,1)[12]$) with drift, which has 2 AR coefficients ($ar1$, $ar2$), 2 MA coefficients ($ma1$, $ma2$), 1 seasonal MA coefficient ($sma1$), 1 seasonal AR coefficient ($sar1$) and 1 drift constant. This gives a total of 7 fitted parameters.

So the number of fitted parameters is: $k = 7$

The AIC penalty is: $2k = 2(7) = 14$

The AICc penalty adds the small-sample correction: $\left[\frac{2k(k+1)}{n-k-1} \right]$

Using ($n = 204$): $\left[\frac{2(7)(8)}{204-7-1} = \frac{112}{196} = 0.571 \right]$

Therefore: AIC penalty = 14, Additional AICc correction = 0.571, Total AICc penalty = 14.571

- d. Fit an exponential smoothing model of your choice to `diabetes.rds`. Use `glance()` to view the log likelihood and AICc values of both this model and the ARIMA model from Q2.

```
fits <- diabetes |>
  model(
    arima = ARIMA(log(Cost)),
    ets    = ETS(log(Cost))
  )

glance(fits)
```

```
# A tibble: 2 x 11
  .model sigma2 log_lik    AIC   AICc    BIC ar_roots  ma_roots      MSE
  <chr>   <dbl>   <dbl>  <dbl>  <dbl>  <dbl> <list>    <list>    <dbl>
1 arima  0.00359   266.  -516.  -515.  -490. <cpl [14]> <cpl [14]> NA
2 ets    0.00366    38.2  -42.5  -39.2   13.9 <NULL>    <NULL>    0.00337
# i 2 more variables: AMSE <dbl>, MAE <dbl>
```

- e. Can we say which method is a better fit to the data by comparing their log likelihood or AICc? Explain why or why not.

Yes, based on the results, the ARIMA model appears to be the better fit to the data when compared using AICc. The ARIMA model has a much lower AICc value than the ETS model: ARIMA : ($AICc = -515.23726$), ETS : ($AICc = -39.18075$)

Since a lower AICc indicates a better balance between goodness of fit and model complexity, this suggests that the ARIMA model is preferred.

The ARIMA model also has a much higher log likelihood: ARIMA : ($\log L = 266.01207$), ETS : ($\log L = 38.23554$)

This indicates that the ARIMA model provides a substantially better in-sample fit.

Therefore, for these fitted models, we would conclude that the ARIMA model is the better-fitting method. x

4. Dynamic regression (adapted from Q10.4 in Hyndman & Athanasopoulos)

This exercise concerns `aus_accommodation`: the total quarterly takings from accommodation and the room occupancy level for hotels, motels, and guest houses in Australia, between January 1998 and June 2016. Total quarterly takings are in millions of Australian dollars.

```
aus_accommodation
```

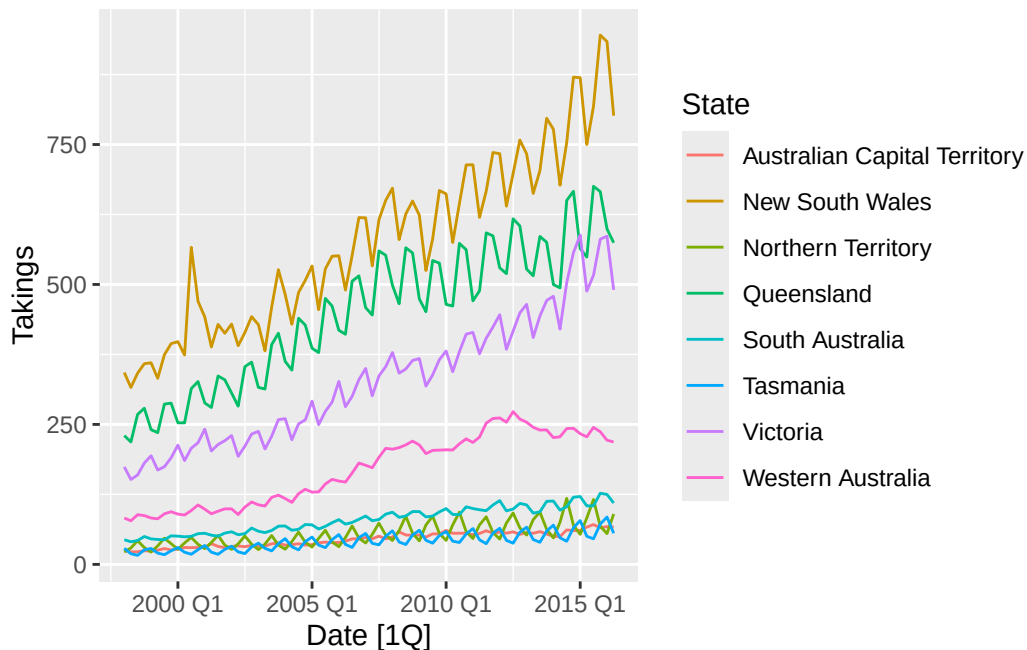
```
# A tsibble: 592 x 5 [1Q]
# Key:      State [8]
  Date State Takings Occupancy CPI
```

	<qtr>	<chr>	<dbl>	<dbl>	<dbl>
1	1998 Q1	Australian Capital Territory	24.3	65	67
2	1998 Q2	Australian Capital Territory	22.3	59	67.4
3	1998 Q3	Australian Capital Territory	22.5	58	67.5
4	1998 Q4	Australian Capital Territory	24.4	59	67.8
5	1999 Q1	Australian Capital Territory	23.7	58	67.8
6	1999 Q2	Australian Capital Territory	25.4	61	68.1
7	1999 Q3	Australian Capital Territory	28.2	66	68.7
8	1999 Q4	Australian Capital Territory	25.8	60	69.1
9	2000 Q1	Australian Capital Territory	27.3	60.9	69.7
10	2000 Q2	Australian Capital Territory	30.1	64.7	70.2

i 582 more rows

```
aus_accommodation |>
  autoplot()
```

Plot variable not specified, automatically selected ``.vars = Takings``



- Perform inflation adjustment for Takings (using the CPI column), creating a new column in the tsibble called `AdjTakings`.

```
aus_acccommodation <- aus_accommodation |>
  mutate(AdjTakings = Takings / CPI * first(CPI))
```

```
aus_acccommodation
```

```
# A tsibble: 592 x 6 [1Q]
```

```
# Key:      State [8]
```

	Date	State	Takings	Occupancy	CPI	AdjTakings
	<qtr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	1998 Q1	Australian Capital Territory	24.3	65	67	24.3
2	1998 Q2	Australian Capital Territory	22.3	59	67.4	22.2
3	1998 Q3	Australian Capital Territory	22.5	58	67.5	22.4
4	1998 Q4	Australian Capital Territory	24.4	59	67.8	24.1
5	1999 Q1	Australian Capital Territory	23.7	58	67.8	23.4
6	1999 Q2	Australian Capital Territory	25.4	61	68.1	25.0
7	1999 Q3	Australian Capital Territory	28.2	66	68.7	27.5
8	1999 Q4	Australian Capital Territory	25.8	60	69.1	25.0
9	2000 Q1	Australian Capital Territory	27.3	60.9	69.7	26.2
10	2000 Q2	Australian Capital Territory	30.1	64.7	70.2	28.7

```
# i 582 more rows
```

- b. For each state, fit a dynamic regression model of AdjTakings with seasonal dummy variables, a piecewise linear time trend with one knot at 2008 Q1, and ARIMA errors.

```
aus_accommodation <- aus_accommodation |>
  mutate(
    AdjTakings = Takings / CPI * first(CPI),
    trend = as.numeric(Date),
    knot = pmax(as.numeric(Date) - as.numeric(yearquarter("2008 Q1")), 0)
  )

fit <- aus_accommodation |>
  model(ARIMA(AdjTakings ~ season() + trend + knot))
```

- c. What model was fitted for the state of Victoria? Does the time series exhibit constant seasonality?

```
fit |> filter(State == "Victoria") |> report()
```

Series: AdjTakings

Model: LM w/ ARIMA(1,0,0)(0,0,1)[4] errors

Coefficients:

	ar1	sma1	season()year2	season()year3	season()year4	trend
	0.6797	0.3765	-38.8346	-24.1782	-8.6832	1.6963
s.e.	0.0855	0.1236	2.6666	3.1030	2.7168	0.0385

```

      knot
      0.9951
s.e.  0.3882

```

```

sigma^2 estimated as 77.4:  log likelihood=-262.93
AIC=541.86   AICc=544.07   BIC=560.29

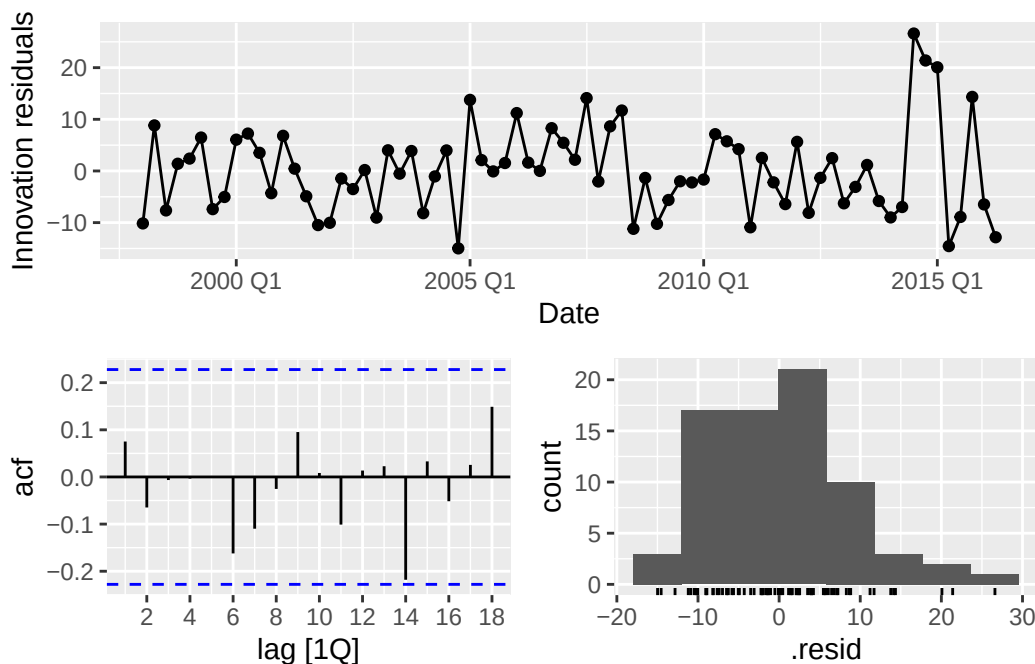
```

The fitted model for Victoria is a linear model with $ARIMA(1,0,0)(0,0,1)[4]$ errors, incorporating seasonal dummies, a piecewise linear trend with a knot at 2008 Q1, an $AR(1)$ term, and a seasonal $MA(1)$ term.

Yes, the time series exhibits constant seasonality under this model. This is because quarterly seasonality is represented using fixed seasonal dummy variables, so the seasonal effects are assumed to remain constant in magnitude over time rather than increasing or decreasing with the level of the series.

- d. Check that the residuals of the model in c) look like white noise.

```
fit |> filter(State == "Victoria") |> gg_tsresiduals()
```



```

fit |> filter(State == "Victoria") |>
  augment() |>
  features(.innov, ljung_box, lag = 8)

```

```
# A tibble: 1 x 4
```

	State	.model	lb_stat	lb_pvalue
	<chr>	<chr>	<dbl>	<dbl>
1	Victoria	ARIMA(AdjTakings ~ season() + trend + knot)	4.01	0.856

The residuals appear to behave like white noise. The residual plot shows random fluctuations around zero with no clear trend or seasonality. The ACF values are mostly within the significance bounds, indicating little remaining autocorrelation.

The Ljung–Box test gave: [$Q^* = 4.0099$, $p\text{-value} = 0.8562$]

Since the p -value is greater than 0.05, we fail to reject the null hypothesis of no autocorrelation.

Therefore, the residuals are consistent with white noise, suggesting that the fitted model for Victoria is adequate.

- e. Use an ARIMA model to forecast CPI, then use these to forecast the (non-adjusted) takings for Victoria until the end of 2017. *Hint: You will have to use the `new_data` argument to `forecast()`.*

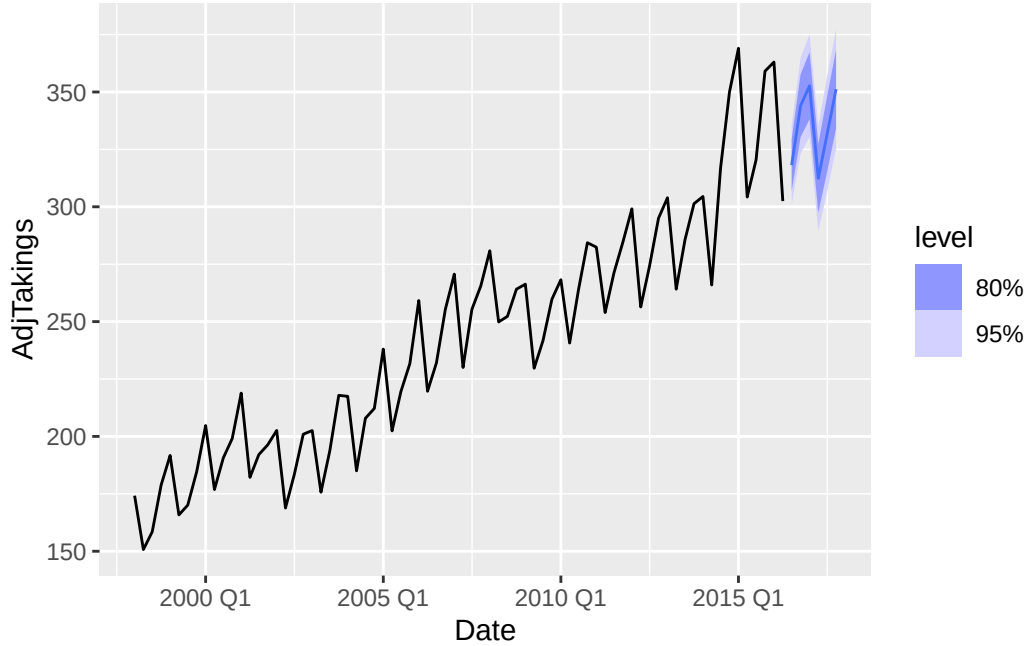
```
fit_cpi <- aus_accommodation |>
  filter(State == "Victoria") |>
  model(ARIMA(CPI))

fc_cpi <- fit_cpi |> forecast(h = 6)

new_data_vic <- new_data(
  aus_accommodation |> filter(State == "Victoria"), 6) |>
  mutate(CPI = fc_cpi$.mean,
         trend = as.numeric(Date),
         knot = pmax(as.numeric(Date) - as.numeric(yearquarter("2008 Q1")), 0))

fc <- fit |>
  filter(State == "Victoria") |>
  forecast(new_data = new_data_vic) |>
  mutate(Takings_forecast = .mean * new_data_vic$CPI / 100)

fc |> autoplot(aus_accommodation |> filter(State == "Victoria"))
```

- f. What sources of uncertainty have not been taken into account in the prediction intervals?

The prediction intervals do not account for all sources of uncertainty. They do not fully include the uncertainty in the CPI forecasts that were used as future explanatory variables, nor the uncertainty from estimating the regression and ARIMA model (knot, trend) parameters. They also ignore the possibility of future structural changes in the tourism industry, model selection uncertainty, and unexpected economic shocks or policy changes. Therefore, the actual forecast uncertainty is likely to be larger than the reported prediction intervals.

5. State space models

Consider the model

$$T_t = T_{t-1} + \epsilon_t$$

$$S_t = 0.8S_{t-4} + \eta_t$$

$$Y_t = T_t + S_t + V_t$$

where $(\epsilon_t) \sim WN(0, \sigma_1^2)$, $\eta_t \sim WN(0, \sigma_2^2)$, and $(V_t) \sim WN(0, \sigma_3^2)$. We observe data $Y_1, \dots, Y_n$.

a. Write the model in the standard (matrix) form of a linear Gaussian state space model.

$$X_t = \begin{pmatrix} T_t \\ S_t \\ S_{t-1} \\ S_{t-2} \\ S_{t-3} \end{pmatrix}$$

$$X_t = F X_{t-1} + W_t$$

$$F = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.8 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

$$W_t = \begin{pmatrix} \epsilon_t \\ \eta_t \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$Q = \begin{pmatrix} \sigma_1^2 & 0 & 0 & 0 & 0 \\ 0 & \sigma_2^2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$Y_t = H X_t + V_t$$

$$H = (1 \quad 1 \quad 0 \quad 0 \quad 0)$$

$$V_t \sim WN(0, \sigma_3^2)$$

b. Does $\lim_{t \rightarrow \infty} \text{Var}(S_t - \hat{S}_{t|n})$ exist? If so, what is its value?

$$S_t = 0.8 S_{t-4} + \eta_t$$

Since

$$|0.8| < 1,$$

the seasonal component is stationary.

Let

$$\text{Var}(S_t) = \gamma.$$

Then

$$\gamma = 0.8^2\gamma + \sigma_2^2$$

$$\gamma(1 - 0.8^2) = \sigma_2^2$$

$$\gamma = \frac{\sigma_2^2}{1 - 0.8^2}$$

Therefore,

$$\begin{aligned}\lim_{t \rightarrow \infty} \text{Var}(S_t - \hat{S}_{t|n}) &= \frac{\sigma_2^2}{1 - 0.8^2} \\ &= \frac{\sigma_2^2}{0.36}\end{aligned}$$

c. Does $\lim_{t \rightarrow \infty} \text{Var}(T_t - \hat{T}_{t|n})$ exist? If so, what is its value?

$$T_t = T_{t-1} + \epsilon_t$$

This is a random walk, so its uncertainty increases over time.

For

$$t > n,$$

we have

$$T_t = T_n + \epsilon_{n+1} + \epsilon_{n+2} + \cdots + \epsilon_t.$$

Therefore, the forecast error variance contains

$$(t - n)\sigma_1^2.$$

As

$$t \rightarrow \infty,$$

this goes to infinity.

Hence,

$$\lim_{t \rightarrow \infty} \text{Var}(T_t - \widehat{T}_{t|n})$$

does not exist as a finite value.

Equivalently,

$$\text{Var}(T_t - \widehat{T}_{t|n}) \rightarrow \infty.$$

AI Declaration:

ChatGPT was used as a supplementary tool to assist with conceptual clarification and LaTeX formatting during the preparation of this assignment. All mathematical derivations, computations, and final answers were carried out by the author.